INCH-POUND
MS27183L
w/Amendment 1
14 October 2022
SUPERSEDING
MS27183L
22 October 2021

DETAIL SPECIFICATION SHEET

WASHER, FLAT (ROUND, STEEL, CADMIUM OR ZINC-NICKEL PLATED) GENERAL PURPOSE

This specification is approved for use by all Departments and Agencies of the Department of Defense.

The requirements for acquiring the product described herein shall consist of this specification sheet and FF-W-92.

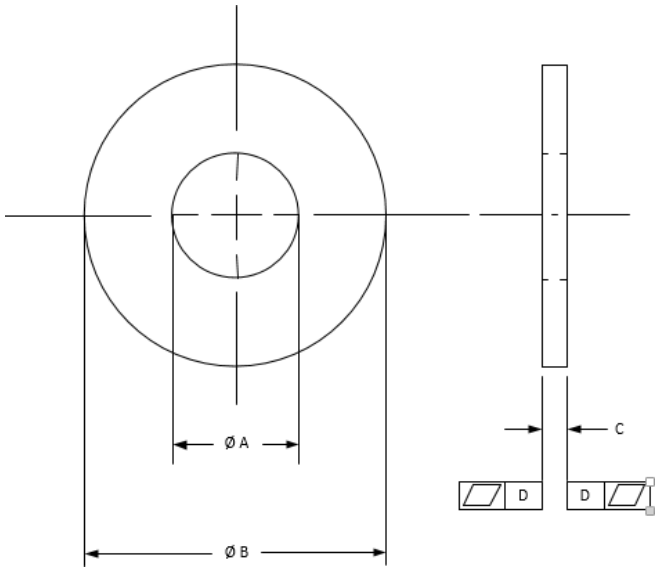


FIGURE 1. Washer, flat.

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TABLE I. Dash numbers and dimensions.

DASH NUMBER CADMIUM PLATED	DASH NUMBER ZN-NI PLATED	A	TOLERANCE (+)	TOLERANCE (-)	B	TOLERANCE (+)	TOLERANCE (-)	C THICKNESS (BASIC)	TOLERANCE MAX	TOLERANCE MIN	D
-1	-1QX	0.078	0.000	0.005	0.188	0.000	0.005	0.020	0.025	0.016	0.005
-2	-2QX	0.094	0.000	0.005	0.250	0.000	0.005	0.020	0.025	0.016	0.005
-3	-3QX	0.125	0.000	0.005	0.250	0.000	0.005	0.022	0.028	0.017	0.005
-4	-4QX	0.125	0.008	0.005	0.312	0.008	0.005	0.032	0.040	0.025	0.005
-5	-5QX	0.156	0.008	0.005	0.312	0.008	0.005	0.035	0.048	0.027	0.005
-43	-43QX	0.156	0.008	0.005	0.250	0.000	0.015	0.016	0.020	0.013	0.005
-6	-6QX	0.156	0.008	0.005	0.375	0.015	0.005	0.049	0.065	0.036	0.005
-7	-7QX	0.188	0.008	0.005	0.375	0.015	0.005	0.049	0.065	0.036	0.005
-41	-41QX	0.188	0.008	0.005	0.438	0.015	0.005	0.049	0.065	0.036	0.005
-8	-8QX	0.219	0.008	0.005	0.438	0.015	0.005	0.049	0.065	0.036	0.005
-42	-42QX	0.219	0.008	0.005	0.500	0.015	0.005	0.049	0.065	0.036	0.005
-44	-44QX	0.125	0.008	0.005	0.375	0.015	0.005	0.032	0.040	0.025	0.005
-45	-45QX	0.141	0.008	0.005	0.281	0.005	0.005	0.032	0.040	0.025	0.005
-46	-46QX	0.188	0.008	0.005	0.500	0.015	0.005	0.065	0.080	0.051	0.005
-47	-47QX	0.219	0.008	0.005	0.365	0.005	0.015	0.032	0.040	0.025	0.005
-48	-48QX	0.250	0.005	0.020	0.938	0.005	0.015	0.065	0.080	0.051	0.005
-49	-49QX	0.250	0.015	0.005	0.562	0.015	0.005	0.032	0.040	0.025	0.005
-50	-50QX	0.250	0.015	0.005	0.625	0.015	0.005	0.032	0.040	0.025	0.005
-9	-09QX	0.250	0.015	0.005	0.562	0.015	0.005	0.065	0.080	0.051	0.005
-10	-10QX	0.281	0.015	0.005	0.625	0.015	0.005	0.065	0.080	0.051	0.005
-51	-51QX	0.250	0.015	0.015	1.000	0.015	0.005	0.065	0.080	0.051	0.005
-52	-52QX	0.281	0.015	0.005	0.734	0.015	0.007	0.065	0.080	0.051	0.005
-11	-11QX	0.312	0.015	0.005	0.734	0.015	0.007	0.065	0.080	0.051	0.005
-12	-12QX	0.344	0.015	0.005	0.688	0.015	0.007	0.065	0.080	0.051	0.005
-13	-13QX	0.375	0.015	0.005	0.875	0.030	0.007	0.083	0.104	0.064	0.005
-53	-53QX	0.312	0.015	0.005	0.875	0.015	0.015	0.065	0.075	0.051	0.005
-54	-54QX	0.312	0.015	0.005	0.500	0.015	0.005	0.035	0.048	0.027	0.005
-55	-55QX	0.375	0.015	0.005	0.734	0.015	0.005	0.083	0.094	0.064	0.005
-56	-56QX	0.375	0.015	0.005	0.625	0.030	0.010	0.035	0.048	0.027	0.005
-57	-57QX	0.406	0.015	0.005	0.875	0.015	0.007	0.063	0.070	0.051	0.005
-58	-58QX	0.412	0.010	0.005	0.625	0.005	0.005	0.032	0.040	0.025	0.005
-14	-14QX	0.406	0.015	0.005	0.812	0.015	0.007	0.065	0.080	0.051	0.005
-15	-15QX	0.438	0.015	0.005	1.000	0.030	0.007	0.083	0.104	0.064	0.005
-59	-59QX	0.438	0.015	0.005	1.365	0.015	0.007	0.083	0.104	0.064	0.005
-60	-60QX	0.438	0.015	0.005	0.875	0.015	0.007	0.065	0.080	0.051	0.005
-61	-61QX	0.438	0.015	0.005	1.625	0.015	0.007	0.065	0.104	0.051	0.005
-62	-62QX	0.469	0.005	0.020	0.734	0.030	0.007	0.065	0.080	0.051	0.005
-16	-16QX	0.469	0.015	0.005	0.922	0.015	0.007	0.065	0.080	0.051	0.005
-17	-17QX	0.500	0.015	0.005	1.250	0.030	0.007	0.083	0.104	0.064	0.005
-18	-18QX	0.531	0.015	0.005	1.062	0.030	0.007	0.095	0.121	0.074	0.005

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TABLE I. Dash numbers and dimensions - Continued.

DASH NUMBER CADMIUM PLATED	DASH NUMBER ZN-NI PLATED	A	TOLERANCE (+)	TOLERANCE (-)	B	TOLERANCE (+)	TOLERANCE (-)	C THICKNESS (BASIC)	TOLERANCE MAX	TOLERANCE MIN	D
-19	-19QX	0.562	0.015	0.005	1.375	0.030	0.007	0.109	0.132	0.086	0.005
-20	-20QX	0.594	0.015	0.005	1.156	0.030	0.007	0.095	0.121	0.074	0.005
-63	-63QX	0.656	0.015	0.010	2.125	0.030	0.007	0.095	0.115	0.080	0.005
-21	-21QX	0.656	0.030	0.007	1.312	0.030	0.007	0.095	0.121	0.074	0.005
-22	-22QX	0.688	0.030	0.007	1.750	0.030	0.007	0.134	0.160	0.108	0.005
-23	-23QX	0.812	0.030	0.007	1.469	0.030	0.007	0.134	0.160	0.108	0.005
-24	-24QX	0.812	0.030	0.007	2.000	0.030	0.007	0.148	0.177	0.122	0.005
-25	-25QX	0.938	0.030	0.007	1.750	0.030	0.007	0.134	0.160	0.108	0.010
-26	-26QX	0.938	0.030	0.007	2.250	0.030	0.007	0.165	0.192	0.136	0.010
-27	-27QX	1.062	0.030	0.007	2.000	0.030	0.007	0.134	0.160	0.108	0.010
-28	-28QX	1.062	0.030	0.007	2.500	0.030	0.007	0.165	0.192	0.136	0.010
-29	-29QX	1.250	0.030	0.007	2.750	0.030	0.007	0.165	0.192	0.136	0.010
-30	-30QX	1.375	0.030	0.007	3.000	0.030	0.007	0.165	0.192	0.136	0.010
-31	-31QX	1.500	0.045	0.010	3.250	0.045	0.010	0.180	0.213	0.153	0.010
-32	-32QX	1.625	0.045	0.010	3.500	0.045	0.010	0.180	0.213	0.153	0.010
-33	-33QX	1.750	0.045	0.010	3.750	0.045	0.010	0.180	0.213	0.153	0.010
-34	-34QX	1.875	0.045	0.010	4.000	0.045	0.010	0.180	0.213	0.153	0.010
-35	-35QX	2.000	0.045	0.010	4.250	0.045	0.010	0.180	0.213	0.153	0.010
-36	-36QX	2.125	0.045	0.010	4.500	0.045	0.010	0.180	0.213	0.153	0.010
-37	-37QX	2.375	0.045	0.010	4.750	0.045	0.010	0.220	0.248	0.193	0.010
-38	-38QX	2.625	0.045	0.010	5.000	0.045	0.010	0.238	0.280	0.210	0.010
-39	-39QX	2.875	0.065	0.010	5.250	0.065	0.010	0.259	0.310	0.228	0.010
-40	-40QX	3.125	0.065	0.010	5.500	0.065	0.010	0.284	0.327	0.249	0.010

REQUIREMENTS

1. MATERIAL, STEEL, CARBON GRADES 1005 THRU 1020 (UNS G10050 THRU UNS G10200), IN ACCORDANCE WITH ASTM A108, ASTM A109/A109M, OR ASTM A1011/A1011M, ANY TEMPER.

2. PROTECTIVE COATING AND SURFACE TREATMENTS: ZINC-NICKEL IN ACCORDANCE WITH MIL-PRF-32647, TYPE K16, OR CADMIUM PLATE /1 IN ACCORDANCE WITH SAE-AMS-QQ-P-416, TYPE II, CLASS 3.

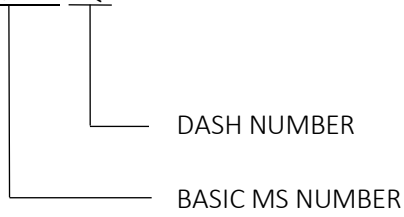
/1 Cadmium is an R1, R2, and D1 hazardous substance in the Aerospace and Defense Declarable Substance List. Zinc-Nickel plate per MIL-PRF-32647 is a suitable replacement with design authority approval.

3. SURFACE TEXTURE, BEARING SURFACE ROUGHNESS SHALL NOT EXCEED 125 MICROINCHES (Ra) IN ACCORDANCE WITH ASME B46.1.

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4. PART NUMBER, THE PART NUMBER SHALL CONSIST OF THE BASIC MS NUMBER FOLLOWED BY A DASH NUMBER FROM TABLE I AND PLATING SUFFIX (IF REQUIRED).

EXAMPLE MS27183-1QX



EXAMPLE PART NUMBERS:

MS27183-1 - WASHER, FLAT (ROUND STEEL), ID - .078, OD - .188, THICKNESS .020, CADMIUM PLATED.

MS27183-1QX - WASHER, FLAT (ROUND STEEL), ID - .078, OD - .188, THICKNESS .020, ZINC-NICKEL PLATED.

TABLE II. PART NUMBERS.

PART NUMBERS				
CANCELLED				PREFERRED
UNCOATED	PHOSPHATE	CAD OR ZN	CADMIUM	Zinc-Nickel
MS15795-1	MS15795-101	MS15795-201	MS27183-1	MS27183-1QX
-2	-102	-202	-2	-2QX
-3	-103	-203	-3	-3QX
-4	-104	-204	-4	-4QX
-5	-105	-205	-5	-5QX
-6	-106	-206	-6	-6QX
-7	-107	-207	-7	-7QX
-8	-108	-208	-8	-8QX
-9	-109	-209	-9	-9QX
-10	-110	-210	-10	-10QX
-11	-111	-211	-11	-11QX
-12	-112	-212	-12	-12QX
-13	-113	-213	-13	-13QX
-14	-114	-214	-14	-14QX
-15	-115	-215	-15	-15QX
-16	-116	-216	-16	-16QX
-17	-117	-217	-17	-17QX
-18	-118	-218	-18	-18QX
-19	-119	-219	-19	-19QX
-20	-120	-220	-20	-20QX
-21	-121	-221	-21	-21QX
-22	-122	-222	-22	-22QX
-23	-123	-223	-23	-23QX
-24	-124	-224	-24	-24QX
-25	-125	-225	-25	-25QX
-26	-126	-226	-26	-26QX
-27	-127	-227	-27	-27QX
-28	-128	-228	-28	-28QX
-29	-129	-229	-29	-29QX
-30	-130	-230	-30	-30QX
-31	-131	-231	-31	-31QX
-32	-132	-232	-32	-32QX
-33	-133	-233	-33	-33QX
-34	-134	-234	-34	-34QX
-35	-135	-235	-35	-35QX
-36	-136	-236	-36	-36QX
-37	-137	-237	-37	-37QX
-38	-138	-238	-38	-38QX
-39	-139	-239	-39	-39QX
-40	-140	-240	-40	-40QX

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NOTES:

1. ALL DIMENSIONS ARE IN INCHES
2. IN THE EVENT OF A CONFLICT BETWEEN THE TEXT OF THIS DOCUMENT AND THE REFERENCES CITED HEREIN, THE TEXT OF THIS DOCUMENT SHALL TAKE PRECEDENCE
3. UNLESS OTHERWISE SPECIFIED, ISSUE OF REFERENCED DOCUMENTS ARE THOSE IN EFFECT AT THE TIME OF SOLICITATION
4. CADMIUM OR ZINC COATED STEEL WASHERS ONLY, PREVIOUSLY SPECIFIED ON MS15795 WERE CANCELLED AND SUPERSEDED BY CADMIUM OR ZINC-NICKEL PLATED STEEL WASHERS SPECIFIED HEREIN
5. USE THE PART NUMBERS GIVEN IN TABLE I OR TABLE II
6. IN THE PRECEDING SHEET OF THIS STANDARD, THE CANCELLED WASHERS CANNOT ALWAYS BE REPLACED WITH LIKE FINISHES
7. EXISTING STOCKS OF WASHERS SHOULD BE USED UNTIL EXISTING STOCK IS DEPLETED
8. USE ZINC-NICKEL WASHERS FOR NEW DESIGN AND REPLACEMENT
9. THE MARGINS OF THIS SPECIFICATION ARE MARKED WITH VERTICAL LINES TO INDICATE WHERE CHANGES FROM THE PREVIOUS ISSUE WERE MADE. THIS WAS DONE AS A CONVENIENCE ONLY AND THE GOVERNMENT ASSUMES NO LIABILITY WHATSOEVER FOR ANY INACCURACIES IN THESE NOTATIONS. BIDDERS AND CONTRACTORS ARE CAUTIONED TO EVALUATE THE REQUIREMENTS OF THIS DOCUMENT BASED ON THE ENTIRE CONTENT IRRESPECTIVE OF THE MARGINAL NOTATIONS AND RELATIONSHIP TO THE PREVIOUS ISSUE

CONCLUDING MATERIAL

CUSTODIANS:
ARMY – AT
NAVY – OS
DLA – CQ

PREPARING ACTIVITY:
ARMY – AT

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